

**Determinants and Dynamics of Crime Intensity at Urban Transit Stops: Linking
Demographic, and Social Risk Factors**

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Abstract

Many people rely on the use of public transit every day. However, with the increasing rates of crime, there remains significant concern as to how safe public transit stations are. While several studies analyze these crime rates, few investigate the factors behind why transit stations have become hotspots for crime. We perform our analysis at bus stop level while using sociodemographic and built environment indicators to better understand crime around transit stations. We use the Washington DC bus system as a case study. Our sociodemographic factors are classified into categories based on race, gender, age, median income, education, employment, and poverty, while our built environment indicators include proximity to fire and police stations and road density. For each variable we created a Poisson regression based on type of crime: assault crime count, theft crime count, and theft vehicle count. Our results reveal that minority races and proximity to police and fire stations have a positive correlation with crime while having a bachelor's degree or higher decreases the likelihood of all types of crime analyzed near transit stops.

Keywords:

Public transit, crime, regression, sociodemographic, environmental

INTRODUCTION

Problem statements and objectives

Public transit is one of the most used transportation systems in the Commonwealth of Virginia with nearly 50% of travelers using bus systems (DRPT, 2024). However, with the increase in the number of people traveling, the amount of crime on buses and around bus stops has increased significantly over the past few years. The Washington Metropolitan Area Transit Authority five-year report shows that buses are the third most frequent spot for crime to occur on public transit (WMATA, 2023). Crime data significantly impacts the use of public transportation. For example, one study in Texas found that the increase in crime had a negative correlation with ridership, meaning that as crime increases, ridership decreases (He & Li, 2024). However, there is a lack of data of what exactly is influencing these higher rates in crime.

In this study, we aim to analyze the sociodemographic and built environment factors that have influenced crime at transit stations. Sociodemographic factors include race, sex, age, median income, education, employment, and poverty, while our built environment indicators include road density, and proximity to fire and police stations. We analyze to see if these factors have any significant relationships to crime around transit stations in the D.C. area. Specifically, we seek to answer this research question: How do localized sociodemographic and built environmental indicators affect crime and the type of in the Washington D.C. area? We base our study using bus stops as our mode of transportation and disaggregated data by the type of crime. The types of crime are assault, theft, and vehicle theft. We use the Washington D.C. area as a case study, highlighting the greater need for research and data collection in public transit crime across the United States.

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LITERATURE REVIEW

Sociodemographic Factors: How They Influence Crime at Transit Stations

Sociodemographic factors can influence crime, especially in densely populated, such as transit stations. Analyzing these factors which could influence crime can help explain why and where these patterns occur. Variables such as sex, age, race, income, employment, and poverty can be strong determinants of crime. This can be seen in areas with high traffic, such as transit stations. High density ones are seen in cities, where there's a high population density in general. To examine the relationship between sociodemographic and crime at transit stations, spatial analysis can be used to reveal these hidden patterns. Mitchell's study (2011) analyzed how socioeconomic and demographic variables influenced property and violent crime in Pittsburgh. Out of the 11 variables analyzed, 3 were found to be significant using hot-spot analysis. Percent of males 14-29 years of age were shown to correlate to property crime, low median household income affected violent crime, and percent unemployed affected both crime types (Mitchell, 2011). This shows that certain sociodemographic factors affect crime and also by type, such as violent or non-violent crime, like property crimes. It also supports researchers' theories that factors such as a lack of employment and economic disadvantages may increase the probability of crime occurring within that area. As well as this study highlights how sociodemographic influence crime rates, it also demonstrates how spatial analysis methods can be useful in revealing these patterns. Spatial analysis lets us see the correlation between these variables that aren't always noticeable. It helps us statistically explain the pattern and relationships between the different variables. Although not transit-specific, this study highlights how sociodemographic does in-fact influence crime and how spatial analysis is an efficient tool to identify these patterns.

Relationships Between Crime and Transit Ridership

Another important factor to consider is how crime interacts with transit stations. Crime can affect an individual's view of public transportation and can sway them to not use it. In urban areas though, public transportation is highly valued and is used by many, as it is most convenient in those high traffic areas. One study evaluates how crime affects ridership at transit stations in Texas, and another study also examines this, but the two different studies are actually contradicting.

In a study conducted by He and Li (2024) a negative regression model was used to assess how crime affects ridership at Texas transit stations. They found that neighborhoods with higher crime rates have lower ridership for their transit stops. This suggests that crime may deter individuals from using public transportation due to these high crime rates, which makes the population fear their safety. This is because this affects the population's perception of unsafety and ultimately has effects on the transit stations they use. Although this shows how crime impacts ridership, it's also crucial to analyze what leads to these crimes that are impacting transit sites.

Conversely, a study by Esfandyari (2020) found that crime has a positive relationship with ridership. This is because it is seen to be associated with high activity. This study defines a theory called the Routine Active Theory, which explains crime by examining the daily routines of offenders and victims. It claims that a crime is more likely to occur by there being a suitable target, a motivated offender, and a lack of a capable guardian. Since this study found that higher crime results from a higher amount of ridership, it supports this theory by suggesting that greater concentrations of people give way for more potential targets, which creates more offenders of crime. So, from this viewpoint, higher ridership does not necessarily reduce crime but may

increase crime due to increased population density. These two conflicting relationships further address why analysis is important to figuring out why these crimes occur at transit sites.

Built Environment Factors and Crime Characteristics

Like sociodemographic factors, environmental factors are also determinants of crime. Specifically, built-environmental factors, such as land use, population density, and infrastructure are good predictors as to what could explain crime in the area, especially transit stations in urban locations due to their highly concentrated population. The study by Loukaitou-Sideris, et.al, (2001) examines how built environment factors such as population density and land use influence crime at transit stations, and Irvin-Erickson and La Vigne (2015) researches the differences between the crimes that occur at different times of day at metro rails.

Loukaitou-Sideris, et.al, (2001) examined the transit at Los Angeles. In their study, crime was aggregated into two crime types: Type 1 and Type 2. Type 1 crime is serious crime, such as homicide, rape, assault, and burglary or theft of vehicles. Type 2 is less serious crime, which was primarily vandalism. Using multiple regression models, Loukaitou-Sideris, et.al, (2001) determined that population density contributed to a higher rate of Type 2 crime (less serious crime). They stated that there was a significant positive correlation between the population density in the vicinity of the station and the Type 2 crime. The r value for crime counts was 0.899 and the r value for crime rate was 0.542 (Loukaitou-Sideris, et.al, 2001). They stated that this seems logical as more vandalism would result in areas with a lot of people. They analyzed a station that has essentially no population in the surrounding area, and the r value dropped to 0.855, showing the population does in fact influence the crime rate, and also the type of crime.

Loukaitou-Sideris, et.al (2001) also examined land use, such as school, bars, liquor stores, pawnshops, and abandoned buildings, in relation to crime. They found that there was a strong positive correlation between liquor stores and Type 2 crime at stations, r equaling 0.815. Also, check cashing establishments and pawnshops were noted to also have high Type 2 crimes. Another interesting fact is that Type 1 and 2 crimes were high at stations surrounded by residential land use and lower at stations with office and industrial land use. This can be due to lower population densities in the office and industrial areas. This study shows that certain built environmental factors have a significant influence on crimes in transit areas and the type of crime it is.

Similarly, Irvin-Erickson and La Vigne (2015) also stratified crime into groups, but three temporal groups. These were peak hours, off-peak day hours, and off-peak night hours. This study used a spatio-temporal regression to evaluate the relationship between crimes at transit stations in D.C., including larceny, aggravated assault, and robbery to the time of day and by different variables, such as SES and environmental. Their results show that the crime type is affected by the time of day. 56% of aggravated assaults occurred during the non-peak night hours, which suggests that crimes like this are likely to happen at times when there's a low population density. On the other hand, 80% of the robberies were divided equally between peak hours and non-peak night hours, with the other 20% being observed during non-peak day hours. This shows that different characteristics of crime, such as the time of day, have a role in analyzing when and where these crimes occur.

These studies reveal how analyzing social and environmental factors correlates with crime, while also adding in other characteristics, such as the type of crime and the time when the

crime occurs. Transit stations have many patterns of human activity, which is why analyzing the transit stations of Washington, D.C. is critical for understanding localized crime patterns.

DATA AND METHOD

In this study, we evaluated how public transportation can affect frequency and type of crime rate in the Washington D.C. area. To achieve this, we collected sociodemographic and built environment data and performed a Poisson regression analysis to better understand the correlation between crime hotspots and their influences, specifically around bus stops.

Crime and Bus Stop Data

Our analysis will be at bus stop level with separate models created for three different crime types. Crime was aggregated by three types: assault, theft, and vehicle theft. This was our independent variable in our Poisson regression analysis to analyze if crime is related to public transportation, specifically bus stops in Washington, D.C. The crime data was also at bus stop level, so bus stop data was also included in the dataset. The bus stop ID, latitude, and longitude were important attributes that we extracted so we could create maps with the D.C. tracts as the background. Since this data was at bus stop level, we merged the other data variables that were at tract level with the bus stop by using the within predicate. This crime and bus stop dataset was created by Armita Kar and obtained from the ArcGIS Online hub.

Sociodemographic and Built Environment Data

We collected social variables including age, race, median income, education, employment, and poverty. Total population was also extracted in order to perform calculations, such as for population density. All variables were imported as percents utilizing the Data Profile tables from the American Community Survey (ACS) for the year 2022 directly from the census library in Python.

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Separate models for different types of crime rates

Commented [SH6]: @Amber Ruth Gamgort were homicide and sexual assault also analyzed? And were they also divided into day, evening, and night? or did we not use those?

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The types of races included were Black, Asian alone, and two or more races. The percent of population of Hispanic origin and not of Hispanic origin were included. Sex was the total percent of males and total percent of females. Age was categorized as percent under 18, percent 18 years and older, and percent 65 years and older. Education was stratified as percent high school with no diploma, percent high school with a diploma (including equivalency), percent bachelor's degree or higher, and percent graduate or professional degree. Median income in dollars, percent employed, percent unemployed, and poverty, which was percentage of families and people whose income in the past 12 months is below the poverty line.

Population density was also a factor we considered when performing our analysis. This variable was obtained by calculating it. We used the total population imported from the ACS, where we divided it by the geometry area in the D.C. tracts. The specific unit the area was in was kilometers squared. The tract area was obtained from the U.S. Census Bureau TIGER/Line shapefile for the year 2022.

Other built environment factors we utilized were road density and distance to police stations from the tract, and distance to fire stations from the tract. Distance to police and fire stations were obtained from the Open Data DC built environment indicators. Road density was calculated, but the roadways were obtained from the District Department of Transportation. This shapefile has seven types of functional classes of roadways, including interstate, other freeway and expressway, principal arterial, minor arterial, collector, minor collector, and local roadways. Due to some similarities between roadways, we decided to group the roadways into three categories: Highway, arterial collector, and local. Highways included interstates and other freeway and expressways, arterial and collector included all arterial and collector lines, and local

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included just local roadways. Road density was calculated by dividing the length of the road types by the area in kilometers squared.

Modeling

A Poisson regression analysis was completed for each of the three types of crime counts which include assault, other theft, and vehicle theft as dependent variables. Independent variables chosen include sociodemographic, environmental, and road. The Poisson regression is used for counting based data unlike a standard linear regression. At first, the model was not very informative. However, after removing certain independent variables, we found the best fit based on a 0.05 confidence interval. Therefore, variables with a significant level of a p-value less than 0.05 were considered statistically significant. The output gave important statistical information including pseudo R², coefficients, and standard error. In addition, coefficient plots were created to show the difference in significance levels between each variable and error associated for each of the three types of crime.

Results

In this study, we examined multiple types of crime occurring at bus stop locations across the Washington, D.C. metropolitan area to assess whether sociodemographic and environmental variables could explain the variation in crime counts across the bus stops. Our independent variables included poverty rate, unemployment rate, race (e.g. Black, Asian, or persons identifying as two or more races), age (under 18), educational attainment (people with bachelor's and people with graduate degrees), road density, and proximity to police and fire stations. Figure

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(1) illustrates the spatial distribution of bus stop locations across the study area - the Washington D.C Metropolitan area.

A Poisson regression model was used to evaluate predictors of assault count, motor vehicle theft count, and other theft count. Additional crime categories, such as homicide and sexual abuse, were excluded due to insufficient counts with the data, which we felt would not provide reliable or meaningful statistical results for the particular model. As a result of this analysis, the model's performance varied across crime types as indicated by pseudo- R^2 values and its coefficients.

The assault model produced a relatively low pseudo- R^2 value (~ 0.125), suggesting that while some predictors are statistically significant, a substantial portion of the variation in assault counts remains unexplained (Figures 2 and Figure 3). Similarly, the motor vehicle theft model produced a pseudo- R^2 value of ~ 0.171 , indicating only moderate variation. However, in contrast to these two independent variables, the model for other types of theft demonstrated a pseudo- R^2 value of ~ 0.378 thus, suggesting our dependent variables better capture variance associated.

For the assault model, road density and proximity to police and fire stations were strong positive predictors, indicating that assault incidents are more likely to occur in highly connected/dense areas as well as more developed areas. Additionally, higher levels of educational attainment (specifically the count of individuals with bachelor's degrees per tract) were associated with decreased assault counts as it revealed a negative coefficient value of -0.0246 , thus suggesting that higher education attainment tends to experience lower levels of assault.

For other theft, proximity to police and fire stations and higher numbers of road density were strongly associated with increased theft counts, thus indicating that theft is more prevalent in areas which have greater connectivity and therefore activity. The higher pseudo- R^2 of 0.3782

for this model suggests that these factors explain much of the variation seen in theft-related crimes (Figures 4 and Figure 5).

Motor vehicle theft displayed a different pattern compared to the other two types of crimes. Poverty, under 18 and person with graduate degrees, were negatively associated with vehicle theft counts, thus suggesting that these types of crimes are not as strongly associated with motor vehicle theft (Figures 6 and Figure 7). For this type of crime, it may have been best to look at other potential factors outside of sociodemographic such as difference in commercial vs residential areas (land use) or levels of traffic. However, similar to the other models, road density and proximity to emergency services were still strong positive predictors in this model. The pseudo- R^2 of 0.1712 indicates that while these variables explain some variation, additional factors likely influence vehicle theft patterns and should be further investigated in future studies.

Overall, the results indicate that crime at bus stops is influenced by a combination of sociodemographic factors, as well as environmental factors. However, given that the pseudo- R^2 values across all three models were low to moderate in value, it is best to assume that additional variables should be considered in future studies to improve the model's performance. Potential factors which could be taken into consideration to improve the model could be factors such as land use characteristics (e.g. commercial versus residential areas) measures of foot traffic, or lighting conditions at the bus stop.

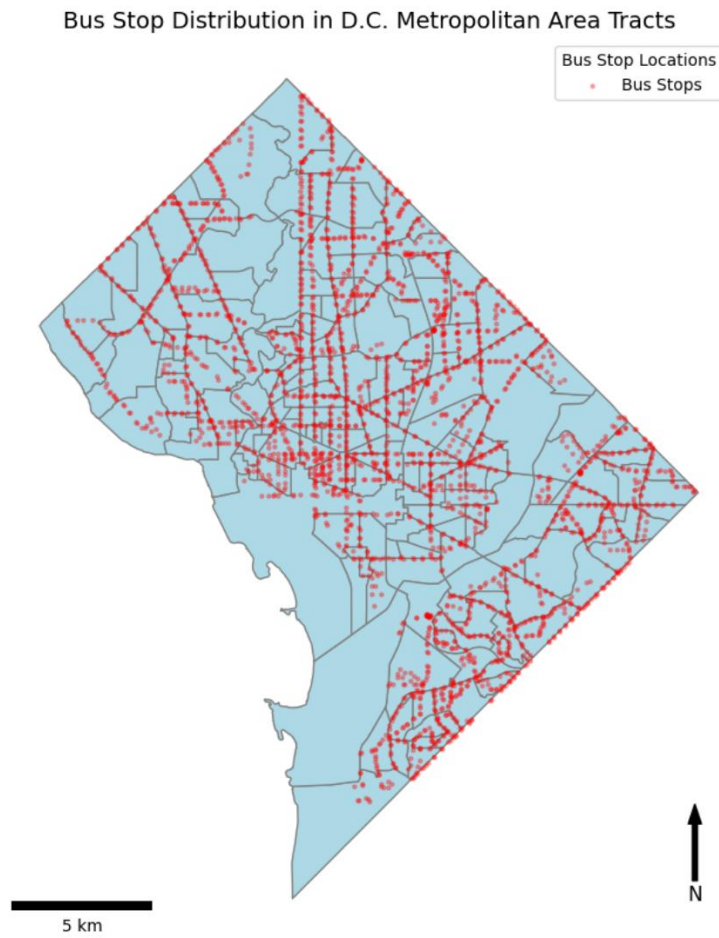


Figure 1: Bus stop locations (red points) across Washington, D.C. metropolitan area census tracts.

Optimization terminated successfully.
Current function value: 1.929706
Iterations: 29
Function evaluations: 37
Gradient evaluations: 37

Poisson Regression Results

Dep. Variable: assault_count

No. Observations: 2401

Model: Poisson

Df Residuals: 2389

Method: MLE

Df Model: 11

Date: Thu, 30 Apr 2026

Pseudo R-squ.: 0.1253

Time: 00:41:18

Log-Likelihood: -4633.2

converged: True

LL-Null: -5297.0

Covariance Type: nonrobust

LLR p-value: 5.354e-278

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.7184	0.222	3.236	0.001	0.283	1.154
pct_poverty	0.0067	0.002	3.456	0.001	0.003	0.011
pct_unemployed	0.0117	0.005	2.591	0.010	0.003	0.021
pct_black	0.0003	0.002	0.172	0.863	-0.003	0.004
pct_asian	0.0302	0.008	3.912	0.000	0.015	0.045
pct_two_or_more	0.0011	0.005	0.234	0.815	-0.008	0.011
pct_under_18	-0.0024	0.003	-0.907	0.364	-0.008	0.003
pct_bachelors_plus	-0.0246	0.003	-9.365	0.000	-0.030	-0.019
pct_grad_degree	0.0032	0.004	0.909	0.363	-0.004	0.010
road_density	0.0617	0.012	5.365	0.000	0.039	0.084
proximity_to_police_station	0.2037	0.043	4.723	0.000	0.119	0.288
proximity_to_fire_station	0.4860	0.060	8.087	0.000	0.368	0.604

Figure 2: Table results of assault count per bus stop using the Poisson regression model.

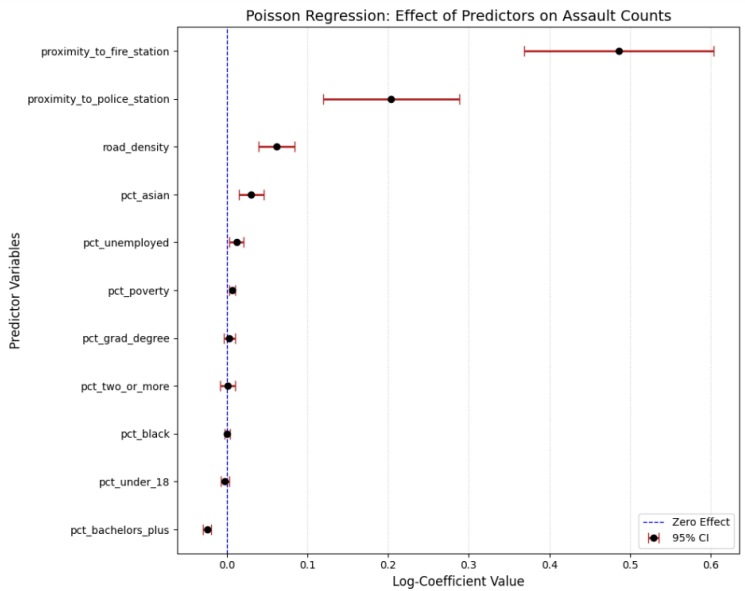


Figure 3: Graphical results of the assault count per bus stop using the Poisson regression model.

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Optimization terminated successfully.
Current function value: 21.056139
Iterations: 25
Function evaluations: 31
Gradient evaluations: 31

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Poisson Regression Results						
Dep. Variable:	other_theft_count	No. Observations:	2401			
Model:	Poisson	Df Residuals:	2389			
Method:	MLE	Df Model:	11			
Date:	Thu, 30 Apr 2026	Pseudo R-squ.:	0.3782			
Time:	00:41:18	Log-Likelihood:	-50556.			
converged:	True	LL-Null:	-81302.			
Covariance Type:	nonrobust	LLR p-value:	0.000			

	coef	std err	z	P> z	[0.025	0.975]
Intercept	2.4473	0.038	64.628	0.000	2.373	2.522
pct_poverty	-0.0001	0.000	-0.266	0.790	-0.001	0.001
pct_unemployed	0.0111	0.001	10.921	0.000	0.009	0.013
pct_black	-7.157e-06	0.000	-0.027	0.979	-0.001	0.001
pct_asian	0.0229	0.001	23.060	0.000	0.021	0.025
pct_two_or_more	0.0333	0.001	49.920	0.000	0.032	0.035
pct_under_18	-0.0119	0.000	-25.981	0.000	-0.013	-0.011
pct_bachelors_plus	0.0176	0.000	40.934	0.000	0.017	0.018
pct_grad_degree	-0.0221	0.000	-50.118	0.000	-0.023	-0.021
road_density	0.0140	0.002	8.701	0.000	0.011	0.017
proximity_to_police_station	0.5863	0.007	82.154	0.000	0.572	0.600
proximity_to_fire_station	1.1358	0.013	87.971	0.000	1.111	1.161

Figure 4: Table results of other theft types count per bus stop using the Poisson regression model.

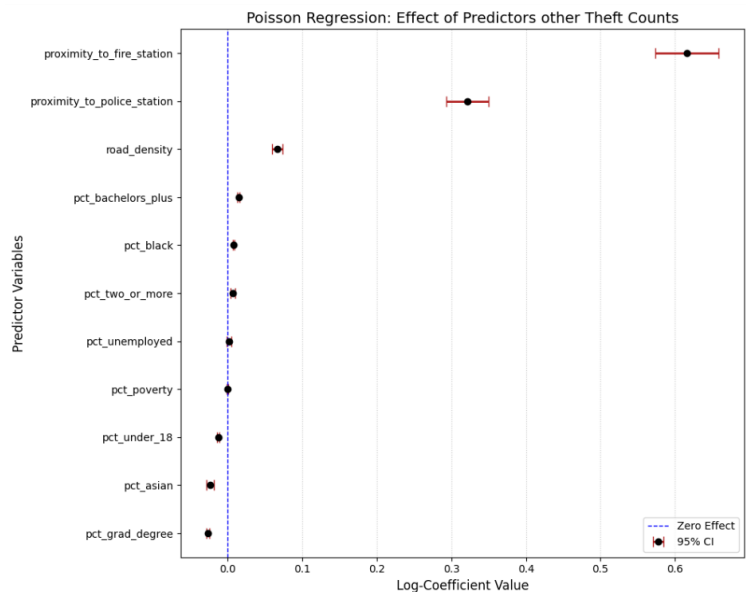


Figure 5: Graphical results of the other theft types count per bus stop using the Poisson regression model.

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    Current function value: 5.868054
    Iterations: 22
    Function evaluations: 26
    Gradient evaluations: 26

===== Poisson Regression Results =====
Dep. Variable:  theft_vehicle_count      No. Observations:      2401
Model:          Poisson                  Df Residuals:          2389
Method:         MLE                      Df Model:              11
Date:           Thu, 30 Apr 2026         Pseudo R-squ.:         0.1712
Time:           00:41:19                  Log-Likelihood:        -14089.
Converged:      True                     LL-Null:               -16999.
Covariance Type: nonrobust                LLR p-value:           0.000

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              coef      std err      z      P>|z|      [0.025      0.975]
-----
Intercept      1.6613      0.071     23.307    0.000      1.522      1.801
pct_poverty    -2.954e-07      0.001     -0.000    1.000     -0.001      0.001
pct_unemployed  0.0022      0.002      1.274    0.203     -0.001      0.005
pct_black       0.0085      0.001     15.816    0.000      0.007      0.010
pct_asian      -0.0231      0.003     -8.874    0.000     -0.028     -0.018
pct_two_or_more  0.0074      0.002      4.722    0.000      0.004      0.011
pct_under_18   -0.0126      0.001    -13.977    0.000     -0.014     -0.011
pct_bachelors_plus  0.0150      0.001     18.171    0.000      0.013      0.017
pct_grad_degree -0.0259      0.001    -24.887    0.000     -0.028     -0.024
road_density    0.0667      0.004     18.819    0.000      0.060      0.074
proximity_to_police_station  0.3218      0.015     22.151    0.000      0.293      0.350
proximity_to_fire_station  0.6158      0.022     28.403    0.000      0.573      0.658
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Figure 6: Table results of the theft types count per bus stop using the Poisson regression model

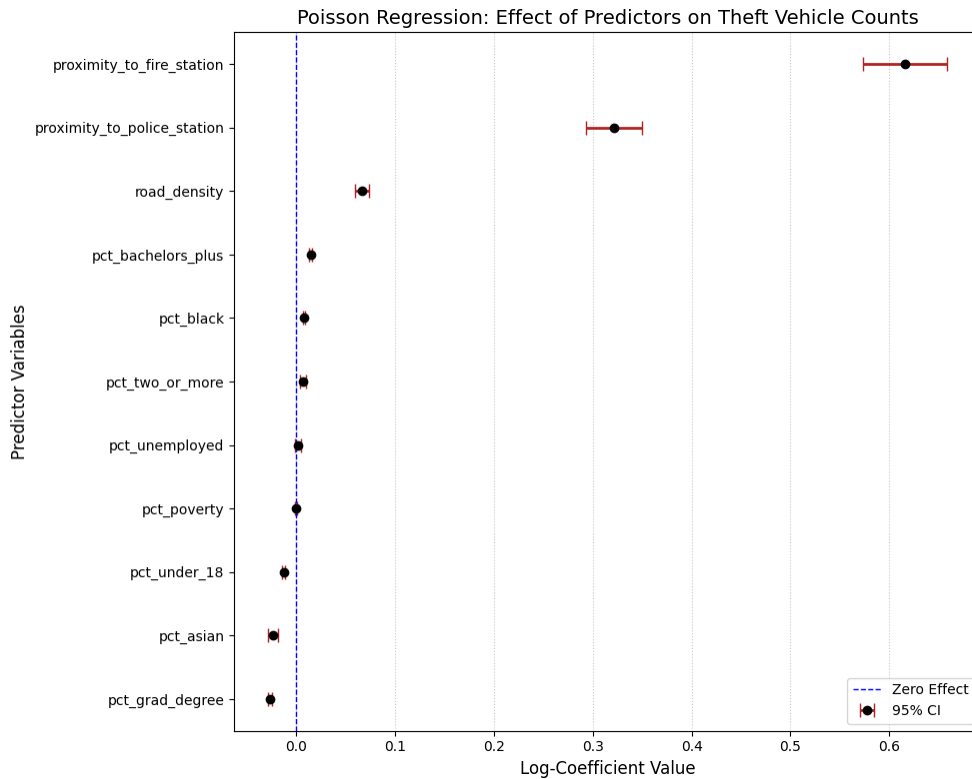


Figure 7: Graphical results of the theft types count per bus stop using the Poisson regression model

Discussion and Conclusion

This study provides important insight into how both sociodemographic and environmental factors influence crime around transit stops in the Washington, D.C. metropolitan area. By analyzing crime rates at the census tract level, we applied a Poisson regression model to three categories of crime such as assault, theft, and motor vehicle theft. The results of this analysis show that variation in these crime types is associated with factors such as road density, proximity to emergency services (e.g., police and fire stations), and education attainment. These findings are significant because public transportation is a critical

asset to urban infrastructure as it supports mobility and accessibility for the community. Understanding the drivers of crime near transit stops is essential, as perceptions of safety directly influence whether individuals choose to use these infrastructures. Ultimately, this analysis could pose the incentive to have more effective urban planning strategies which in turn can provide better safety and accessibility in transit environments for all individuals.

The results suggest that activity and connectivity play a more prominent role in crime patterns rather than sociodemographic factors alone. The positive relationship between road density and proximity to emergency services indicates that crime is more likely to occur in highly connected and high-activity areas where there is greater interaction among people, and movement in general. In contrast, the negative relationship between higher education attainment levels and crime suggests that education may reduce the likelihood of certain crimes, particularly assault to occur. Motor vehicle theft exhibited a different pattern compared to the other crime types, with weaker or model outputs. This suggests that vehicle theft may be influenced more strongly by other factors, such as land use patterns or traffic flow, rather than the variables considered in this model's study.

These findings have several implications for urban planning and public safety. Transit planners can use these results from our factors such as road density and connectivity to identify high-risk areas and prioritize safety interventions, including improved lighting, surveillance, and infrastructure design (such as police monitoring locations). From a policing perspective, resources can be allocated more effectively by focusing on areas with high levels of activity rather than higher crime rates. From a fire and emergency services standpoint, identifying locations with higher crime rates, particularly those associated with road density and connectivity factors, they can support more strategic planning and risk mitigation. Fire

departments can use this information to anticipate areas with elevated levels of incidents or emergencies. Additionally, understanding patterns of accessibility and connectivity enables emergency responders to optimize routing, improve response times, and ensure adequate service coverage in high-demand areas which also have high crime rates. Additionally, the observed relationship between education and reduced crime suggests that long-term investments in education and community resources may contribute to safer transit environments. Overall, the results support the idea that crime prevention near transit should focus not only on enforcement from police, but also on infrastructure design and patterns of higher levels of activity and movement.

Despite these contributions that our model produced, several opportunities for improvement should be acknowledged. Certain crime categories, such as homicide and sexual abuse, were excluded due to a high number of zero values within the dataset, which limited their suitability for a Poisson regression model analysis. Additionally, some dependent variables were removed due to instability in the model (i.e. coefficient values reaching max values), which may have reduced the overall explanatory power these variables could have shown. While the model produced meaningful results, the exclusion of these variables suggests that important factors influencing crime may not have been fully captured and therefore should be used in future runs of this study. Future analyses which explore these variables or include different combinations of said variables, as well as possible alternative models besides the Poisson model could improve model performance and provide a more comprehensive understanding of the crime dynamics which occur near transit stops.

Future research should incorporate additional variables to better understand factors contributing to crime in transit environments. For example, land use data (i.e. commercial

versus residential areas) may help explain differences across crime types in more specific areas. Measures of foot traffic or population mobility may be a good variable as it could provide a more direct representation of human activity which could better represent areas which could have more crime rates. Finally, separating the crime occurrence data to represent time of day or day of the week could reveal important patterns and therefore provide more answers to the variation we see in the model. By incorporating these suggestions to future studies, improvements to the model can be made which can then turn provide a more complete understanding of transit-related crime dynamics.

Overall, this study provided further insight as to how sociodemographic and built environment factors can influence crime around public transportation sites in the Washington, D.C metropolitan area. As urbanized areas heavily rely on public transportation to provide citizens access to mobility, it is important to analyze the implications which could improve these infrastructures. By performing this analysis, we hope that our findings can encourage policy makers to implement strategies which will reduce crime rates in these bus stop areas. These applications will make transit stations in D.C. safer, which will increase the overall populations' perception of safety, thus making D.C. a safer and low crime area, especially in transit areas.

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